

Lesson 1 · Homework

NCEA Level 2 · AS 91262 · Gradients, differentiation and tangents

Time: about 60 minutes

Answer version · tutor's working, not an official mark scheme

Show all working and write $f'(x)$ each time. [A] Achieved, [M] Merit, [E] Excellence.

1. Differentiate.

[A]

(a) $f(x) = 4x^5 - 2x^3 + 9$

$$f'(x) = 20x^4 - 6x^2$$

(b) $y = 7 - 3x + 5x^2$

$$\frac{dy}{dx} = -3 + 10x$$

(c) $g(x) = \frac{1}{2}x^4 - \frac{1}{3}x^3 + x$

$$g'(x) = 2x^3 - x^2 + 1$$

(d) $y = (x + 4)(x - 1)$

$$y = x^2 + 3x - 4$$

$$\frac{dy}{dx} = 2x + 3$$

2. Differentiate $y = x^2(3x - 5)$.

[A]

$$y = 3x^3 - 5x^2$$

$$\frac{dy}{dx} = 9x^2 - 10x$$

3. Differentiate $f(x) = (2x - 3)^2$.

[A]

$$f(x) = 4x^2 - 12x + 9$$

$$f'(x) = 8x - 12$$

4. Find the gradient of $f(x) = x^3 + 2x^2 - 5x + 1$ at the point where $x = 2$.

[A]

$$f'(x) = 3x^2 + 4x - 5$$

$$f'(2) = 12 + 8 - 5 = 15$$

5. Find the gradient of $y = 4 - 3x^2$ at the point where $x = -2$. [A]

$$\frac{dy}{dx} = -6x; \text{ at } x = -2: -6(-2) = 12$$

6. For $y = x^4 - 8x$, find the gradient at $x = 1$. Is the curve going up or down there? [A]

$$\frac{dy}{dx} = 4x^3 - 8; \text{ at } x = 1: 4 - 8 = -4$$

Negative gradient, so the curve is going down at $x = 1$.

7. Find the coordinates of the point on $y = x^2 - 6x + 10$ where the gradient is 4. [A]

$$2x - 6 = 4, \text{ so } x = 5$$

$$y = 25 - 30 + 10 = 5; \text{ the point is } (5, 5)$$

8. Find the x -values where $f(x) = x^3 - 3x^2 - 9x + 2$ has gradient 0. [A]

$$f'(x) = 3x^2 - 6x - 9 = 0$$

$$x^2 - 2x - 3 = 0, \text{ so } (x - 3)(x + 1) = 0$$

$$x = 3 \text{ or } x = -1$$

9. Find the coordinates of the two points on $y = x^3 - 12x + 1$ where the gradient is 15. [A]

$$3x^2 - 12 = 15, \text{ so } x^2 = 9, x = \pm 3$$

$$x = 3: y = 27 - 36 + 1 = -8; \quad x = -3: y = -27 + 36 + 1 = 10$$

Points $(3, -8)$ and $(-3, 10)$

10. The curve $f(x) = ax^3 + 2x^2 - 1$ has gradient 32 at $x = 2$. Find a . [A]

$$f'(x) = 3ax^2 + 4x$$

$$f'(2) = 12a + 8 = 32, \text{ so } a = 2$$

11. Find the equation of the tangent to $y = 3x^2 - x + 2$ at $x = 1$. [A]

$$\text{Point: } y = 3 - 1 + 2 = 4, \text{ so } (1, 4)$$

$$\frac{dy}{dx} = 6x - 1, \text{ so } m = 5$$

$$y - 4 = 5(x - 1), \text{ so } y = 5x - 1$$

12. Find the equation of the tangent to $f(x) = x^3 - 4x$ at $x = -1$. [A]

Point: $f(-1) = -1 + 4 = 3$, so $(-1, 3)$

$f'(x) = 3x^2 - 4$, so $m = 3 - 4 = -1$

$y - 3 = -1(x + 1)$, so $y = -x + 2$

13. The curve $y = px^2 + qx$ passes through $(2, 2)$, and its gradient at that point is 5. Find p and q . [M]

Point: $4p + 2q = 2$

Gradient: $\frac{dy}{dx} = 2px + q$, so $4p + q = 5$

Subtract: $q = -3$; then $4p = 8$, $p = 2$

14. The tangent to $y = x^2 + 2x - 3$ at the point P is parallel to the line $y = 6x - 1$. Find the coordinates of P and the equation of the tangent. [M]

Parallel means gradient 6: $2x + 2 = 6$, so $x = 2$

$y = 4 + 4 - 3 = 5$, so P is $(2, 5)$

$y - 5 = 6(x - 2)$, so $y = 6x - 7$

15. **Extension** The tangent to $y = x^2$ at the point where $x = a$ passes through $(0, -9)$. Find the possible values of a and the equations of these tangents. [E]

Point (a, a^2) ; gradient $2a$; tangent: $y - a^2 = 2a(x - a)$, so $y = 2ax - a^2$

Through $(0, -9)$: $-9 = 0 - a^2$, so $a^2 = 9$, $a = 3$ or $a = -3$

$a = 3$: $y = 6x - 9$; $a = -3$: $y = -6x - 9$

Optional extra: StudyTime Level 2 Calculus Guide, "Stop and Check" on pages 10 and 12.